NAME: _____ Geometry (FL B.E.S.T.)

3.3 Rotations

A rotation turns every point of a figure through the same angle about a fixed center, which is why every image point stays exactly as far from the center as its preimage did. Beyond the three memorized rules for rotating about the origin (90°, 180°, 270°), this section extends rotation to a center that is *not* the origin (translate the center to the origin, rotate, translate back), works backward to identify an unknown rotation angle from a point and its image, and previews how repeated rotations connect to rotational symmetry in Section 3.5.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Rotate $(3, 2)$ $90°$ CCW about origin.

2. Rotate $(5, -1)$ $180°$ about origin.

3. Rotate $(1, 4)$ $270°$ CCW about origin.

4. Rotate $(2, 3)$ $90°$ CW about origin.

5. $(x, y) \to$?

6. $(x, y) \to$?

7. Rotation reverse orientation?

8. Is rotation a rigid motion?

9. Rotate $(-2, 5)$ $90°$ CCW.

10. Rotate $(-3, -6)$ $180°$.

11. $270°$ CCW equals how many degrees CW?

12. Does the center of rotation move?

13. Figure congruent to its rotate?

14. Rotate $(6, 4)$ $90°$ CCW about the center $(3, 4)$ (not the origin).

15. A point $(4, -1)$ maps to $(1, 4)$ under a rotation about the origin. Which rotation is it?

✓ Answer Key

1. $(-2, 3)$
2. $(-5, 1)$
3. $(4, -1)$
4. $(3, -2)$
5. $(-y, x)$
6. $(-x, -y)$
7. No
8. Yes
9. $(-5, -2)$
10. $(3, 6)$
11. 90° CW
12. No
13. Yes
14. $(3, 7)$
15. 90° counterclockwise (rule $(x, y) \to (-y, x)$: $(4, -1) \to (1, 4)$)